

# Amrita Raghunath

Machine Learning Researcher — Computer Vision, Multimodal Sequence Modeling, Sign Language & Accessibility AI

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## Research Summary

Machine learning researcher with 6+ years building multimodal deep learning systems for real-time sign language recognition and accessibility. Contributed to a 2025 *IEEE Access* publication on continuous Indian Sign Language (ISL) recognition (WER 19), combining domain-specific pre-training, RGB-pose fusion, and CTC-based sequence modeling; built a government-funded ISL chatbot deployed for public-service accessibility. Seeking research/PhD positions extending multimodal sequence modeling to new decoding problems (e.g., gesture-to-text, brain-to-text).

## Education

### Master of Technology (MTech), Artificial Intelligence

Amrita Vishwa Vidyapeetham

2019 – 2021

Kollam, India

• CGPA: 8.51/10 | Thesis Advisor: Prof. Geetha M | Focus: Algorithms, Pattern Recognition

### Bachelor of Technology (BTech), Computer Science

Amrita Vishwa Vidyapeetham

2014 – 2018

Kollam, India

• CGPA: 8.53/10 | Project Advisor: Prof. Krishnaprasad TR | Focus: Data Science, SciPy  
• Exchange Semester, University of Paderborn, Germany (2018) — Thesis Advisor: Prof. Holger Karl

## Publications

- M. Geetha, N. Aloysius, D. A. Somasundaran, **A. Raghunath**, and P. Nedungadi, "Toward Real-Time Recognition of Continuous Indian Sign Language: A Multi-Modal Approach Using RGB and Pose," *IEEE Access*, vol. 13, pp. 60270–60283, 2025. doi: 10.1109/ACCESS.2025.3554618
  - Proposed **SignFlow**: domain-specific pre-training (3D CNN on isolated ISL videos; Transformer on MediaPipe pose/hand estimates) with dynamic downsampling (6 fps) and CTC loss
  - Achieved Word Error Rate (WER) of 19 on continuous ISL recognition, with a novel detection-rate metric for keyword identification

## Research Experience

### Independent Researcher

Self-directed

Mar 2024 – Present

Kerala, India

- Finalized and published the SignFlow continuous ISL recognition paper in *IEEE Access* (2025)
- Scoping a new research direction in sequence-to-text decoding from non-visual signal sources, informed by recent brain-to-text decoding approaches (e.g., Brain2Qwerty-style architectures), as a generalization of the RGB/pose-to-text pipeline developed in prior work

### Machine Learning Engineer

Amrita Center for Research in Analytics, Technologies & Education (CREATE)

Jan 2022 – Mar 2024

Kollam, India

- Led development of a Conformer-based continuous ISL recognition model in PyTorch, improving on a prior CNN-Transformer baseline through hyperparameter optimization and refined evaluation metrics
- Designed and implemented a CNN-Transformer continuous sign recognition pipeline (isolated-to-continuous transfer pretraining), incorporating MediaPipe/OpenCV-based pose generation and data augmentation
- Built an FAQ-retrieval module for ISL query handling using clustering algorithms and a custom question-answer mapping algorithm, integrated into a government-funded ISL chatbot for public-service accessibility
- Constructed a benchmark static ISL dataset (alphabets, digits 0–9) with MediaPipe/OpenCV preprocessing; benchmarked VGG and ResNet classifiers in PyTorch

### Research Associate

Amrita Centre for Cyber Security Systems and Networks

Mar 2021 – Nov 2021

Kollam, India

- Contributed to applied ML research within the center's cyber-security-oriented research program

### Junior Research Fellow

Amrita Centre for Cyber Security Systems and Networks

Dec 2018 – Jul 2019

Kollam, India

- Developed an unsupervised neural machine translation system for Mizo language using semi-supervised training (BERT encoder, LSTM decoder) over monolingual Mizo/Nagamese corpora
- Automated monolingual corpus collection via Selenium web crawling; trained FastText, GloVe, and MUSE word embeddings

## Selected Earlier Projects

- **Semantic-Guided Scene Description for the Visually Impaired** (2021) — Image captioning model fusing visual-semantic information via Faster-RCNN, BERT, LSTM, and Transformer decoders
- **Real-Time Face Detection for Crowd Surveillance** (2019) — YOLO-based face detection/recognition for CCTV using PixelLib/OpenCV

## Teaching & Mentorship

- **Teaching Assistant**, Object-Oriented Programming (Python), BTech program, Amrita Vishwa Vidyapeetham (2020–2021), under Prof. Neetha J.

- **ISL Accessibility Workshop** (2023) — Workshop for the ISL community with CDAC, Trivandrum, demonstrating the ISL chatbot
- **Leadership**, Ayudh National Leadership Summit (2018) — Mentored 30+ participants

## Technical Skills

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- **Languages:** Python, C++, C, Java, JavaScript, Bash, MATLAB
- **ML/DL Frameworks:** PyTorch, TensorFlow, Keras, HuggingFace, Scikit-learn
- **CV & Sequence Modeling:** Computer Vision, NLP, Generative Models, Reinforcement Learning, MediaPipe, OpenCV
- **Data & Infrastructure:** NumPy, Pandas, PostgreSQL, MySQL, AWS, Docker, Git, Jupyter, Google Colab, Neptune.ai
- **Other:** LaTeX, ImgLab, Linux

## Certifications & Languages

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- **Certification:** Deep Learning Specialization, Coursera (May 2020)
- **Languages:** English (TOEFL 100/120), German (A2), Malayalam (Native), Sanskrit (Conversational)